Classnotes Problems RE, REC, RL

CS & IT ENGINEERING



Theory of Computation

REGULAR LANGUAGE

DPP – 01 Discussion Notes



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[MCQ]

#Q. Which of the following language is non-regular?

A $L = \{wxw^R \mid x, w \in \{a, b\}^*\}$.

B $L = \{wxw \mid w, x \in \{a, b\}^*\}$.

C $L = \{wxwx \mid w, x \in \{a, b\}^+\}$.

D None of these.

C

[MCQ]

#Q. Consider the following grammar G_1 and G_2:

G_1: $S \rightarrow aAb$

$A \rightarrow aB \mid \in$

$B \rightarrow Ab$

G_2: $S \rightarrow aABb$

$A \rightarrow aA \mid \in$

$B \rightarrow bB \mid \in$

Which of the following grammar generates a regular language?

A G_1 only

B G_2 only

C Both G_1 and G_2

D None of these

[NAT]

#Q. Consider the following three languages:

(1) $L = \{a^{n^n} \mid n \geq 1\}$

(2) $L = \{a^{m^n} \mid m = n^2, n \geq 1\}$

(3) $L = \{a^{m^n} \mid n \geq 1, m > n\}$

Total number of regular languages is/are_____.

[MCQ]

#Q. Consider the following grammar

G:

$G = S \rightarrow AB \mid CD$ $\Rightarrow (aa)^*b^* + (aa)^*b^* \Rightarrow (aa)^*b^*$

$A \rightarrow aaA \mid \in$ $\Rightarrow (aa)^*$

$B \rightarrow bB \mid \in$ $\Rightarrow b^*$

$C \rightarrow aaC \mid \in$ $\Rightarrow (aa)^*$

$D \rightarrow bD \mid \in$ $\Rightarrow b^*$

ANY Lang if DFA, NFA, RE, RG

B

bD, bbD, bbbD

The language generated by above grammar is:

A. Finite ✗

B. Infinite but regular ✓

C. Non-regular ✗

D. None of these

[MCQ]

#Q. Which of the following language is non-regular?

A. $L = \{a^{2m} b^n b^n \mid m, n \geq 1\}$

B. $L = \{a^m b^n X \mid m, n \geq 1, X \in \{a,b\}^*\}$

C. $L = \left\{\{a^{n^2}\}^* \mid n \geq 0\right\}$

D. None of these.

[MSQ]

#Q. Regular expression can be used in:

A. Lexical Analysis

B. Pattern matching

C. String matching

D. Syntax analysis

[MCQ]

#Q. Consider the regular expression:

regular expression = $a^*b(a + ba^*)^*$

Above regular expression is equivalent to which of the following below regular expression?

A $ba^*(bb)^*$

B $ba^*(a + ba^*b)^*$

C $(b + aa^*b) + (b + aa^*b)\ (ba^*b + a)\ (ba^*b + a)^*$

D $a^*b\ (a + b)^*$

[MCQ]

#Q. Which of the following statement will generate finite language?



A. PDA with finite stack.

B. Regular expression without kleene star and kleene plus.

C. Regular expression with unary alphabet.

D. Regular expression with binary alphabet.

[MCQ]

#Q. Consider following regular expressions:

[I] $(ab)^*a = a(ab)^*$

[II] $(bb)^*b^* = b^*$

[III] $(b + \in)^+ = b^*$

Which of the following is correct?

A. II and III only.

B. I and II only.

C. All are correct.

D. None of these are correct.

[20]

[MCQ]

#Q. For $L = \{a^n b^m \mid n,m \geq 0\}$
What will be the regular expression ?

A $(a^*b^*)^*$

B (a^*b^*)

C $(ab)^*$

D (b^*a^*)

[MCQ]

#Q. Consider the following regular expressions:

(I) $(aa + aaa)^* = aa^+$

(II) $(a^*b\,(a + b)^* + (a^*b^*)^*) = (a + b)^*$

(III) $(\varepsilon + aaa\,(aaa)^*)\,(\varepsilon + a + aa) = (a + aa + aaa)^*$

Which the following is correct?

A. (I) and (III) only.

B. (II) and (III) only.

C. All are correct.

D. None of these are correct.

[MSQ]

#Q. Which of the following is/are regular expression for the language:
L = {Containing ab as a substring}

A. b* aa*b (a*b*)*

B. (a + b)* (ab)* (a + b)*

C. (a* b*)* ab (a* + b*)*

D. (a + b)* ab (a + b)*

[MCQ]

#Q. What will be the regular expression for $L = \{a^{2n} | n \geq 15\}$ over $\Sigma = \{a\}$

$\{a^{2\cdot 15}\} = a^{30}, a^{32}, a^{34} \ldots$

$= a^{30}(aa)^*$

$a^{30} \cdot \epsilon = a^{30}$

$= a^{30} aa = a^{32}$

A $a^{15}(aa)^*$ X

B $(aa)^* a^{15}$ X

C $a^{30}(aa)^*$ ✓

D None of these

[MCQ]

#Q. Which of the following string does not belong to (ab*)*?

A aaabbaa

B baaaabb

C aaabbb

D ababa

14 Q

All Ques come from CN

WT, DPP, PYQ → 90% CN

THANK - YOU

Revision, PCM,